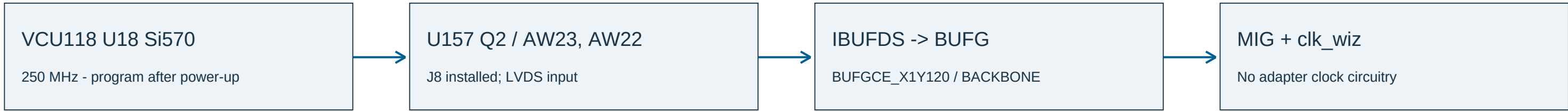


FMC DDR3 无源转接板 - 连线核对图 Rev C

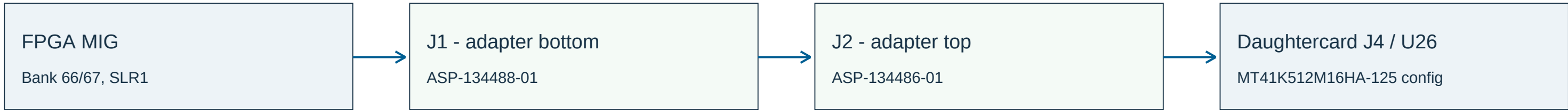
J1 bottom mates VCU118 J2 | J2 top mates daughtercard J4 | Primary handoff: connections.md

母板参考输入：不经过转接板



U18 上电默认 156.25 MHz；J8 开路选 300 MHz。每次断电重启后将 SCUI Si570_0 设为 250 MHz。MIG / XDC 均保持 250 MHz，约束不会给外部器件自动编程。

DRAM 时钟和总线：由 FPGA 输出，经两个 FMC 接口重排



DDR_CK_P: BC9 -> carrier J2.H4 -> adapter J1.H4 / J2.D11 -> daughter J4.D11 -> U26.J7

DDR_CK_N: BC8 -> carrier J2.H5 -> adapter J1.H5 / J2.D12 -> daughter J4.D12 -> U26.K7

CK = 800 MHz / 1600 MT/s. Never connect DDR CK to the 250 MHz reference input.

图纸使用全局同名网络：相同 DDR_x / 3V3 / VADJ / GND 标签在所有页互连。NC 用叉号表示，彼此不互连。第 2-6 页列出两个接口全部 800 针；第 7-8 页是 50 条 DDR 的球号、接点和子板网络核对表。

无晶振、无 PLL、无电平转换器。电源/地接通，其他接点不直通。母板 G6/G7 已停用；子板 G6/G7 分别是 ODT/RESET_n。复位 DNP 下拉及 RN28 条件详见第 9 页。

本次：原始连线 / 静态引脚核对；未改 Vivado / 时钟，未运行综合、实现、物理 DRC 或上板测试。

接插件逐针原理图 - A / B 行

Identical global net labels connect across sheets; NC crosses are independent, not a common net. Electrical pin numbers only.

J1A (BOTTOM)

to VCU118 J2

A1	GND
A2	NC
A3	NC
A4	GND
A5	GND
A6	NC
A7	NC
A8	GND
A9	GND
A10	NC
A11	NC
A12	GND
A13	GND
A14	NC
A15	NC
A16	GND
A17	GND
A18	NC
A19	NC
A20	GND
A21	GND
A22	NC
A23	NC
A24	GND
A25	GND
A26	NC
A27	NC
A28	GND
A29	GND
A30	NC
A31	NC
A32	GND
A33	GND
A34	NC
A35	NC
A36	GND
A37	GND
A38	NC
A39	NC
A40	GND

J2A (TOP)

to daughtercard J4

A1	GND
A2	NC
A3	NC
A4	GND
A5	GND
A6	NC
A7	NC
A8	GND
A9	GND
A10	NC
A11	NC
A12	GND
A13	GND
A14	NC
A15	NC
A16	GND
A17	GND
A18	NC
A19	NC
A20	GND
A21	GND
A22	NC
A23	NC
A24	GND
A25	GND
A26	NC
A27	NC
A28	GND
A29	GND
A30	NC
A31	NC
A32	GND
A33	GND
A34	NC
A35	NC
A36	GND
A37	GND
A38	NC
A39	NC
A40	GND

J1B (BOTTOM)

to VCU118 J2

B1	NC
B2	GND
B3	GND
B4	NC
B5	NC
B6	GND
B7	GND
B8	NC
B9	NC
B10	GND
B11	GND
B12	NC
B13	NC
B14	GND
B15	GND
B16	NC
B17	NC
B18	GND
B19	GND
B20	NC
B21	NC
B22	GND
B23	GND
B24	NC
B25	NC
B26	GND
B27	GND
B28	NC
B29	NC
B30	GND
B31	GND
B32	NC
B33	NC
B34	GND
B35	GND
B36	NC
B37	NC
B38	GND
B39	GND
B40	NC

J2B (TOP)

to daughtercard J4

B1	NC
B2	GND
B3	GND
B4	NC
B5	NC
B6	GND
B7	GND
B8	NC
B9	NC
B10	GND
B11	GND
B12	NC
B13	NC
B14	GND
B15	GND
B16	NC
B17	NC
B18	GND
B19	GND
B20	NC
B21	NC
B22	GND
B23	GND
B24	NC
B25	NC
B26	GND
B27	GND
B28	NC
B29	NC
B30	GND
B31	GND
B32	NC
B33	NC
B34	GND
B35	GND
B36	NC
B37	NC
B38	GND
B39	GND
B40	NC

C35/C37, D1: NC both sides. J1.G6/G7: NC. J2.B2/F2: GND. J2.G17/G26/H6: NC. J1.H2: GND.

接插件逐针原理图 - C / D 行

Identical global net labels connect across sheets; NC crosses are independent, not a common net. Electrical pin numbers only.

J1C (BOTTOM)

to VCU118 J2

C1	GND	
C2	× NC	
C3	× NC	
C4	GND	
C5	GND	
C6	× NC	
C7	× NC	
C8	GND	
C9	GND	
C10	DDR_A6	J2.D15
C11	DDR_A7	J2.D14
C12	GND	
C13	GND	
C14	DDR_CS_N	J2.G19
C15	DDR_CKE	J2.G18
C16	GND	
C17	GND	
C18	× NC	
C19	× NC	
C20	GND	
C21	GND	
C22	× NC	
C23	× NC	
C24	GND	
C25	GND	
C26	DDR_DM1	J2.C23
C27	× NC	
C28	GND	
C29	GND	
C30	× NC	
C31	× NC	
C32	GND	
C33	GND	
C34	GND	
C35	× NC	
C36	GND	
C37	× NC	
C38	GND	
C39	3V3	
C40	GND	

J2C (TOP)

to daughtercard J4

C1	GND	
C2	× NC	
C3	× NC	
C4	GND	
C5	GND	
C6	× NC	
C7	× NC	
C8	GND	
C9	GND	
C10	DDR_A13	J1.G16
C11	DDR_A12	J1.G15
C12	GND	
C13	GND	
C14	DDR_BA0	J1.H13
C15	DDR_BA1	J1.H14
C16	GND	
C17	GND	
C18	DDR_RAS_N	J1.D14
C19	DDR_CAS_N	J1.H17
C20	GND	
C21	GND	
C22	× NC	
C23	DDR_DM1	J1.C26
C24	GND	
C25	GND	
C26	DDR_DQS1_P	J1.H37
C27	DDR_DQS1_N	J1.H38
C28	GND	
C29	GND	
C30	× NC	
C31	× NC	
C32	GND	
C33	GND	
C34	× NC	
C35	× NC	
C36	GND	
C37	× NC	
C38	GND	
C39	3V3	
C40	GND	

J1D (BOTTOM)

to VCU118 J2

D1	× NC	
D2	GND	
D3	GND	
D4	× NC	
D5	× NC	
D6	GND	
D7	GND	
D8	× NC	
D9	× NC	
D10	GND	
D11	DDR_A8	J2.H7
D12	DDR_A9	J2.H8
D13	GND	
D14	DDR_RAS_N	J2.C18
D15	DDR_WE_N	J2.G15
D16	GND	
D17	× NC	
D18	× NC	
D19	GND	
D20	× NC	
D21	× NC	
D22	GND	
D23	× NC	
D24	× NC	
D25	GND	
D26	DDR_DQ8	J2.G36
D27	DDR_DQ9	J2.G37
D28	GND	
D29	× NC	
D30	× NC	
D31	× NC	
D32	3V3	
D33	× NC	
D34	× NC	
D35	GND	
D36	3V3	
D37	GND	
D38	3V3	
D39	GND	
D40	3V3	

J2D (TOP)

to daughtercard J4

D1	× NC	
D2	GND	
D3	GND	
D4	× NC	
D5	× NC	
D6	GND	
D7	GND	
D8	DDR_A1	J1.G10
D9	DDR_A0	J1.G9
D10	GND	
D11	DDR_CK_P	J1.H4
D12	DDR_CK_N	J1.H5
D13	GND	
D14	DDR_A7	J1.C11
D15	DDR_A6	J1.C10
D16	GND	
D17	× NC	
D18	× NC	
D19	GND	
D20	× NC	
D21	× NC	
D22	GND	
D23	× NC	
D24	× NC	
D25	GND	
D26	DDR_DQS0_P	J1.H31
D27	DDR_DQS0_N	J1.H32
D28	GND	
D29	× NC	
D30	× NC	
D31	× NC	
D32	3V3	
D33	× NC	
D34	× NC	
D35	× NC	
D36	3V3	
D37	GND	
D38	3V3	
D39	GND	
D40	3V3	

C35/C37, D1: NC both sides. J1.G6/G7: NC. J2.B2/F2: GND. J2.G17/G26/H6: NC. J1.H2: GND.

接插件逐针原理图 - E / F 行

Identical global net labels connect across sheets; NC crosses are independent, not a common net. Electrical pin numbers only.

J1E (BOTTOM)

to VCU118 J2

E1	GND
E2	NC
E3	NC
E4	GND
E5	GND
E6	NC
E7	NC
E8	GND
E9	NC
E10	NC
E11	GND
E12	NC
E13	NC
E14	GND
E15	NC
E16	NC
E17	GND
E18	NC
E19	NC
E20	GND
E21	NC
E22	NC
E23	GND
E24	NC
E25	NC
E26	GND
E27	NC
E28	NC
E29	GND
E30	NC
E31	NC
E32	GND
E33	NC
E34	NC
E35	GND
E36	NC
E37	NC
E38	GND
E39	VADJ
E40	GND

J2E (TOP)

to daughtercard J4

E1	GND
E2	NC
E3	NC
E4	GND
E5	GND
E6	NC
E7	NC
E8	GND
E9	NC
E10	NC
E11	GND
E12	NC
E13	NC
E14	GND
E15	NC
E16	NC
E17	GND
E18	NC
E19	NC
E20	GND
E21	NC
E22	NC
E23	GND
E24	NC
E25	NC
E26	GND
E27	NC
E28	NC
E29	GND
E30	NC
E31	NC
E32	GND
E33	NC
E34	NC
E35	GND
E36	NC
E37	NC
E38	GND
E39	VADJ
E40	GND

J1F (BOTTOM)

to VCU118 J2

F1	NC
F2	GND
F3	GND
F4	NC
F5	NC
F6	GND
F7	NC
F8	NC
F9	GND
F10	NC
F11	NC
F12	GND
F13	NC
F14	NC
F15	GND
F16	NC
F17	NC
F18	GND
F19	NC
F20	NC
F21	GND
F22	NC
F23	NC
F24	GND
F25	NC
F26	NC
F27	GND
F28	NC
F29	NC
F30	GND
F31	NC
F32	NC
F33	GND
F34	NC
F35	NC
F36	GND
F37	NC
F38	NC
F39	GND
F40	VADJ

J2F (TOP)

to daughtercard J4

F1	NC
F2	GND
F3	GND
F4	NC
F5	NC
F6	GND
F7	NC
F8	NC
F9	GND
F10	NC
F11	NC
F12	GND
F13	NC
F14	NC
F15	GND
F16	NC
F17	NC
F18	GND
F19	NC
F20	NC
F21	GND
F22	NC
F23	NC
F24	GND
F25	NC
F26	NC
F27	GND
F28	NC
F29	NC
F30	GND
F31	NC
F32	NC
F33	GND
F34	NC
F35	NC
F36	GND
F37	NC
F38	NC
F39	GND
F40	VADJ

C35/C37, D1: NC both sides. J1.G6/G7: NC. J2.B2/F2: GND. J2.G17/G26/H6: NC. J1.H2: GND.

接插件逐针原理图 - G / H 行

Identical global net labels connect across sheets; NC crosses are independent, not a common net. Electrical pin numbers only.

J1G (BOTTOM)

to VCU118 J2

G1	GND	
G2	× NC	
G3	× NC	
G4	GND	
G5	GND	
G6	× NC	
G7	× NC	
G8	GND	
G9	DDR_A0	J2.D9
G10	DDR_A1	J2.D8
G11	GND	
G12	DDR_A10	J2.H11
G13	DDR_A11	J2.H10
G14	GND	
G15	DDR_A12	J2.C11
G16	DDR_A13	J2.C10
G17	GND	
G18	DDR_ODT	J2.G6
G19	DDR_RESET_N	J2.G7
G20	GND	
G21	DDR_DQ2	J2.H31
G22	DDR_DQ3	J2.H32
G23	GND	
G24	DDR_DQ4	J2.G34
G25	DDR_DQ5	J2.G33
G26	GND	
G27	DDR_DM0	J2.H28
G28	× NC	
G29	GND	
G30	× NC	
G31	× NC	
G32	GND	
G33	DDR_DQ12	J2.G22
G34	DDR_DQ13	J2.G24
G35	GND	
G36	DDR_DQ14	J2.G25
G37	DDR_DQ15	J2.H29
G38	GND	
G39	VADJ	
G40	GND	

J2G (TOP)

to daughtercard J4

G1	GND	
G2	× NC	
G3	× NC	
G4	GND	
G5	GND	
G6	DDR_ODT	J1.G18
G7	DDR_RESET_N	J1.G19
G8	GND	
G9	DDR_A2	J1.H10
G10	DDR_A3	J1.H11
G11	GND	
G12	DDR_A15	J1.H20
G13	DDR_A14	J1.H19
G14	GND	
G15	DDR_WE_N	J1.D15
G16	DDR_BA2	J1.H16
G17	× NC	
G18	DDR_CKE	J1.C15
G19	DDR_CS_N	J1.C14
G20	GND	
G21	× NC	
G22	DDR_DQ12	J1.G33
G23	GND	
G24	DDR_DQ13	J1.G34
G25	DDR_DQ14	J1.G36
G26	× NC	
G27	× NC	
G28	× NC	
G29	GND	
G30	DDR_DQ0	J1.H22
G31	DDR_DQ1	J1.H23
G32	GND	
G33	DDR_DQ5	J1.G25
G34	DDR_DQ4	J1.G24
G35	GND	
G36	DDR_DQ8	J1.D26
G37	DDR_DQ9	J1.D27
G38	GND	
G39	VADJ	
G40	GND	

J1H (BOTTOM)

to VCU118 J2

H1	× NC	
H2	GND	
H3	GND	
H4	DDR_CK_P	J2.D11
H5	DDR_CK_N	J2.D12
H6	GND	
H7	DDR_A4	J2.H14
H8	DDR_A5	J2.H13
H9	GND	
H10	DDR_A2	J2.G9
H11	DDR_A3	J2.G10
H12	GND	
H13	DDR_BA0	J2.C14
H14	DDR_BA1	J2.C15
H15	GND	
H16	DDR_BA2	J2.G16
H17	DDR_CAS_N	J2.C19
H18	GND	
H19	DDR_A14	J2.G13
H20	DDR_A15	J2.G12
H21	GND	
H22	DDR_DQ0	J2.G30
H23	DDR_DQ1	J2.G31
H24	GND	
H25	DDR_DQ6	J2.H34
H26	DDR_DQ7	J2.H35
H27	GND	
H28	× NC	
H29	× NC	
H30	GND	
H31	DDR_DQS0_P	J2.D26
H32	DDR_DQS0_N	J2.D27
H33	GND	
H34	DDR_DQ10	J2.H38
H35	DDR_DQ11	J2.H37
H36	GND	
H37	DDR_DQS1_P	J2.C26
H38	DDR_DQS1_N	J2.C27
H39	GND	
H40	VADJ	

J2H (TOP)

to daughtercard J4

H1	× NC	
H2	GND	
H3	GND	
H4	× NC	
H5	× NC	
H6	× NC	
H7	DDR_A8	J1.D11
H8	DDR_A9	J1.D12
H9	GND	
H10	DDR_A11	J1.G13
H11	DDR_A10	J1.G12
H12	GND	
H13	DDR_A5	J1.H8
H14	DDR_A4	J1.H7
H15	GND	
H16	× NC	
H17	× NC	
H18	GND	
H19	× NC	
H20	× NC	
H21	GND	
H22	× NC	
H23	× NC	
H24	GND	
H25	× NC	
H26	× NC	
H27	GND	
H28	DDR_DM0	J1.G27
H29	DDR_DQ15	J1.G37
H30	GND	
H31	DDR_DQ2	J1.G21
H32	DDR_DQ3	J1.G22
H33	GND	
H34	DDR_DQ6	J1.H25
H35	DDR_DQ7	J1.H26
H36	GND	
H37	DDR_DQ11	J1.H35
H38	DDR_DQ10	J1.H34
H39	GND	
H40	VADJ	

C35/C37, D1: NC both sides. J1.G6/G7: NC. J2.B2/F2: GND. J2.G17/G26/H6: NC. J1.H2: GND.

接插件逐针原理图 - J / K 行

Identical global net labels connect across sheets; NC crosses are independent, not a common net. Electrical pin numbers only.

J1J (BOTTOM)

to VCU118 J2

J1	GND
J2	× NC
J3	× NC
J4	GND
J5	GND
J6	× NC
J7	× NC
J8	GND
J9	× NC
J10	× NC
J11	GND
J12	× NC
J13	× NC
J14	GND
J15	× NC
J16	× NC
J17	GND
J18	× NC
J19	× NC
J20	GND
J21	× NC
J22	× NC
J23	GND
J24	× NC
J25	× NC
J26	GND
J27	× NC
J28	× NC
J29	GND
J30	× NC
J31	× NC
J32	GND
J33	× NC
J34	× NC
J35	GND
J36	× NC
J37	× NC
J38	GND
J39	× NC
J40	GND

J2J (TOP)

to daughtercard J4

J1	GND
J2	× NC
J3	× NC
J4	GND
J5	GND
J6	× NC
J7	× NC
J8	GND
J9	× NC
J10	× NC
J11	GND
J12	× NC
J13	× NC
J14	GND
J15	× NC
J16	× NC
J17	GND
J18	× NC
J19	× NC
J20	GND
J21	× NC
J22	× NC
J23	GND
J24	× NC
J25	× NC
J26	GND
J27	× NC
J28	× NC
J29	GND
J30	× NC
J31	× NC
J32	GND
J33	× NC
J34	× NC
J35	GND
J36	× NC
J37	× NC
J38	GND
J39	× NC
J40	GND

J1K (BOTTOM)

to VCU118 J2

K1	× NC
K2	GND
K3	GND
K4	× NC
K5	× NC
K6	GND
K7	× NC
K8	× NC
K9	GND
K10	× NC
K11	× NC
K12	GND
K13	× NC
K14	× NC
K15	GND
K16	× NC
K17	× NC
K18	GND
K19	× NC
K20	× NC
K21	GND
K22	× NC
K23	× NC
K24	GND
K25	× NC
K26	× NC
K27	GND
K28	× NC
K29	× NC
K30	GND
K31	× NC
K32	× NC
K33	GND
K34	× NC
K35	× NC
K36	GND
K37	× NC
K38	× NC
K39	GND
K40	× NC

J2K (TOP)

to daughtercard J4

K1	× NC
K2	GND
K3	GND
K4	× NC
K5	× NC
K6	GND
K7	× NC
K8	× NC
K9	GND
K10	× NC
K11	× NC
K12	GND
K13	× NC
K14	× NC
K15	GND
K16	× NC
K17	× NC
K18	GND
K19	× NC
K20	× NC
K21	GND
K22	× NC
K23	× NC
K24	GND
K25	× NC
K26	× NC
K27	GND
K28	× NC
K29	× NC
K30	GND
K31	× NC
K32	× NC
K33	GND
K34	× NC
K35	× NC
K36	GND
K37	× NC
K38	× NC
K39	GND
K40	× NC

C35/C37, D1: NC both sides. J1.G6/G7: NC. J2.B2/F2: GND. J2.G17/G26/H6: NC. J1.H2: GND.

DDR 地址、控制及 CK 对照表 – 28 根

Adapter J1 = carrier J2 contacts; adapter J2 = daughtercard J4 contacts. FPGA ball is not FMC contact.

Adapter global net	J1 / carrier J2	J2 / daughter J4	FPGA ball	Bank	U26 ball	Daughtercard original net
DDR_A0	G9	D9	BD12	66	N3	GPIOR_N_31
DDR_A1	G10	D8	BE12	66	P7	GPIOR_P_31_PLLIN1
DDR_A2	H10	G9	BF12	66	P3	GPIOR_P_28
DDR_A3	H11	G10	BF11	66	N2	GPIOR_N_28
DDR_A4	H7	H14	BC11	66	P8	GPIOR_N_26_CLK9_N
DDR_A5	H8	H13	BD11	66	P2	GPIOR_P_26_CLK9_P
DDR_A6	C10	D15	BD13	66	R8	GPIOR_N_25_CLK10_N
DDR_A7	C11	D14	BE13	66	R2	GPIOR_P_25_CLK10_P
DDR_A8	D11	H7	BE14	66	T8	GPIOR_P_24_CLK11_P
DDR_A9	D12	H8	BF14	66	R3	GPIOR_N_24_CLK11_N
DDR_A10	G12	H11	BE15	66	L7	GPIOR_N_23_CLK12_N
DDR_A11	G13	H10	BF15	66	R7	GPIOR_P_23_CLK12_P
DDR_A12	G15	C11	BC14	66	N7	GPIOR_N_22_CLK13_N
DDR_A13	G16	C10	BC13	66	T3	GPIOR_P_22_CLK13_P
DDR_A14	H19	G13	BB16	66	T7	GPIOR_N_21_CLK14_N
DDR_A15	H20	G12	BC16	66	M7	GPIOR_P_21_CLK14_P
DDR_BA0	H13	C14	BC15	66	M2	GPIOR_P_20_CLK15_P
DDR_BA1	H14	C15	BD15	66	N8	GPIOR_N_20_CLK15_N
DDR_BA2	H16	G16	BA16	66	M3	GPIOR_N_19
DDR_RAS_N	D14	C18	BA14	66	J3	GPIOR_P_18
DDR_CAS_N	H17	C19	BA15	66	K3	GPIOR_N_18
DDR_WE_N	D15	G15	BB14	66	L3	GPIOR_P_19
DDR_CS_N	C14	G19	BB13	66	L2	GPIOR_N_17
DDR_CKE	C15	G18	BB12	66	K9	GPIOR_P_17
DDR_ODT	G18	G6	AV9	66	K1	GPIOR_P_16_PLLIN1
DDR_RESET_N	G19	G7	AV8	66	T2	GPIOR_N_16
DDR_CK_P	H4	D11	BC9	66	J7	GPIOR_P_27_CLK8_P
DDR_CK_N	H5	D12	BC8	66	K7	GPIOR_N_27_CLK8_N

ADDR/BA/control/CK: Bank 66. CK is FPGA output. RESET_n termination/population condition: sheet 9.

DDR 数据字节对照表 – 22 根

Keep DQ/DM/DQS byte grouping and differential polarity. No data bit swapping implied by GPIO names.

Adapter global net	J1 / carrier J2	J2 / daughter J4	FPGA ball	Bank	U26 ball	Daughtercard original net
DDR_DQ0	H22	G30	AW12	67	E3	GPIOB_P_34
DDR_DQ1	H23	G31	AY12	67	F7	GPIOB_N_34
DDR_DQ2	G21	H31	AW11	67	F2	GPIOB_P_33_CDI31
DDR_DQ3	G22	H32	AY10	67	F8	GPIOB_N_33_CDI30
DDR_DQ4	G24	G34	AW13	67	H3	GPIOB_N_32_CDI29
DDR_DQ5	G25	G33	AY13	67	H8	GPIOB_P_32_CDI28
DDR_DQ6	H25	H34	AU11	67	G2	GPIOB_P_31_CDI27
DDR_DQ7	H26	H35	AV11	67	H7	GPIOB_N_31_CDI26
DDR_DQ8	D26	G36	AK15	67	D7	GPIOB_P_30_CDI25
DDR_DQ9	D27	G37	AL15	67	C3	GPIOB_N_30_CDI24
DDR_DQ10	H34	H38	AK12	67	C8	GPIOB_N_29_CDI23
DDR_DQ11	H35	H37	AL12	67	C2	GPIOB_P_29_CDI22
DDR_DQ12	G33	G22	AM13	67	A7	GPIOB_N_28_CDI20
DDR_DQ13	G34	G24	AM12	67	A2	GPIOB_P_27_CDI19
DDR_DQ14	G36	G25	AK14	67	B8	GPIOB_N_27_CDI18
DDR_DQ15	G37	H29	AK13	67	A3	GPIOB_N_26_CDI17
DDR_DM0	G27	H28	AT12	67	E7	GPIOB_P_26_CDI16
DDR_DM1	C26	C23	AL14	67	D3	GPIOB_N_23_CDI12
DDR_DQS0_P	H31	D26	AV10	67	F3	GPIOB_P_25_CDI15
DDR_DQS0_N	H32	D27	AW10	67	G3	GPIOB_N_25_CDI14
DDR_DQS1_P	H37	C26	AJ13	67	C7	GPIOB_P_24_EXTFB
DDR_DQS1_N	H38	C27	AJ12	67	B7	GPIOB_N_24_CDI13

低字节 DQ0..7 / DM0 / DQS0：Bank 67 T0。高字节 DQ8..15 / DM1 / DQS1：Bank 67 T3。原始 GPIO 名 P/N 对单端数据没有差分含义；禁止依名字重新交换数据位。CK / DQS 正负极性保持本表。

电源、地、NC、可选复位下拉与 BOM

No adapter active components. Verify actual daughtercard regulator, decoupling and termination population.

公共电源 / 地：全部同名针加入同一网，不接地针一对一独立接线

J1 C39,D32,D36,D38,D40	3V3	J2 C39,D32,D36,D38,D40
J1 E39,F40,G39,H40	VADJ = 1.5 V	J2 E39,F40,G39,H40
J1 162 contacts	GND	J2 157 contacts

母板真实 VADJ/VCC066/67 必须 1.5 V；共享 FMC Bank 的 C910 JTAG 仍是 LVCMOS18，另行解决电平兼容。子板 VDD/VDDQ=1.5 V，VTT/VREF=0.75 V。子板本地电源/跳线见 README，禁止并联不同来源。

可选上电复位下拉 – 默认不焊（DNP），先处理子板既有 RESET_n 终端



子板 RN28 的 RESET_n / GPIOR_N_16 有 39 ohm 到 VTT 支路。若焊装，4.7k 下拉不能可靠拉低（约 0.744 V）。确认该支路缺省 / 单独隔离后才可焊 R_RESET；不整颗拆掉 RN28，其他 ODT/A13/A9 支路仍需核对。

强制隔离 / 不直通

双方 C35/C37 NC（母板 12 V / 子板 NC，不是地）；双方 D1 NC（PGOOD / 3V3_B）。J1 G6/G7 NC。J2 B2/F2 接 GND；J2 G17/G26/H6 NC。J1 C34/D35 接 GND，子板同号 NC。J1 H2 接地表示插卡。其他未用 I/O / MGT / I2C / 管理 JTAG 均 NC。

最小 BOM

J1 ASP-134488-01 x1 bottom | J2 ASP-134486-01 x1 top | R_RESET 4.7 kohm x1 DNP

不加晶振、PLL 或缓冲芯片。电源去耦按 PDN / 实物需求决定，不在未知负载下编造参数。机械/屏蔽端子根据原厂封装接地，机械孔不默认视为电气针。图纸不是确认子板实物焊装完成的证明。

PCB 制作指导 – 全链路预算，非仅转接板等长

800 MHz CK / 1600 MT/s target is conditional on complete three-board channel SI and timing verification.

1 机械及层叠

上下各一个原厂 FMC；按料号 3D 模型核对高度、孔、定位柱、散热器净空和 A1。不要镜像针号，不同网络顶底焊盘不能共用孔。建议 8 层：SIG/GND/SIG/GND/POWER/SIG/GND/SIG；实际介质、铜厚、线宽由制造厂求解。

2 连续回流与换层

CK/DQS 差分；DQ/DM 单端，按字节分组，同字节除逃线尽量同层。参考地连续、不跨缝；差分同步换层/过孔数一致。回流地过孔距信号换层过孔 <=50 mil（1.27 mm）。不加长测试支路。

3 阻抗不是统一 100 ohm

UG583 数据参考主干单端 39 ohm +/-10%、DQS 76 ohm +/-10%，逃线 50/86 ohm。实际三板已有阻抗和子板 20/39/51/100 ohm 终端须纳入模型，再定 CK/CA/DQ/DQS 网类及制造阻抗券。参考值不是投板许可。

4 补偿总延迟

T总=T封装+T母板+T两次接口+T转接板+T子板。FPGA package min/max 用中值。取得既有板逐线长度/层叠/过孔与连接器模型，再补偿本板；不编造毫米等长值。

UG583 保守全链路参考：实际速度等级/速率可依据官方降额表重审，不能凭空放宽

DQ/DM to own DQS	+/-10 ps
CK P/N and DQS P/N	2 ps each
CA/control to CK (RESET_n excluded)	+/-8 ps
CK to DQS	-149..1796 ps, per guide definition
Data total including FPGA package	<=1186 ps

数据参考最大 PCB 过孔数 2 是全链路，不是每块板 2 个；三板两接口的偏离须做完整 SI。VADJ/3V3 分网、12 V 隔离；电源铜宽/过孔按真实负载、压降、额定值、温升计算。Vivado DRC 不能证明此链路稳定校准。

验证记录、投板检查与资料来源

Revision C: original PDF pin/wire audit + static consistency checks only. No Vivado project changes.

已完成 / 未完成

本次从原始 PDF 独立核对母板 50 对针号/网络、子板全部 400 针引线 与地总线及 50 对针号/GPIO 网络。DDR 位序未变；纠正子板 B2/F2 为 GND、G17/G26/H6 为 NC。800 针网表与 wrapper/XDC 一致；U26 球号、CK/DQS、供电和 RN28 已作图面复查。

本次仅修改连线资料和图纸，未修改 Vivado/XDC/时钟，未综合、实现、编程 U18 或上板测试。版本 B 的参考路径尚未做新物理 DRC；旧 validation 报告仅属于 A。未运行原生 EDA ERC；本图不是 PCB 制造放行。

投板及装配放行清单

原理图 ERC -> 800 接点网表比对 -> PCB DRC / 机械 3D 对接 -> 全链路延迟 / SI -> 制造输出。资料包括 Gerber/ODB++、钻孔、层叠/阻抗券、板框/孔/公差、BOM/坐标/装配图、电测网表与表面处理。当前仅原理图指导，无已布线 PCB / Gerber。

断电验收：50 条 DDR、公共电源地、每个 NC 隔离和相邻针短路。先实测 VADJ=1.5 V、DRAM 电源及复位低电平，再确认 J8 + U18=250 MHz；最后观察 MIG init_calib_complete，做全地址、跨字节、长时间读写和温度测试。

核对来源（完整链接与跳线表见 README）

- VCU118 Rev2.0 schematic: sheets 11/12, 39/40/41, 44; official XDC; native connector GND symbol.
- Efinix fmc-ddr-gpio-card-schematics-v1.0(1).pdf: pages 3/4, actual exterior pin numbers.
- AMD PG150 DDR3 Pin Rules / cross-bank BUFG clocking; VCU118 UG1224 U18/J8/Q2.
- AMD UG583 DDR3 routing constraints, data topology and general memory routing guidelines.
- Samtec ASP-134486-01 / ASP-134488-01 drawings and mating/3D models: verify before PCB release.

Netlist: 800 contacts / 53 connected nets / J1 GND 162, NC 179 / J2 GND 157, NC 184

pinmap.json SHA256: 547dab68aece1769af5dbc1c9737dd6a5960d390cb0e9d6366763e9585d5347f

用 EDA 逐针网表原则与 PDF 渲染核对整理；不是已验证的 EasyEDA 原生工程或已投板 PCB。